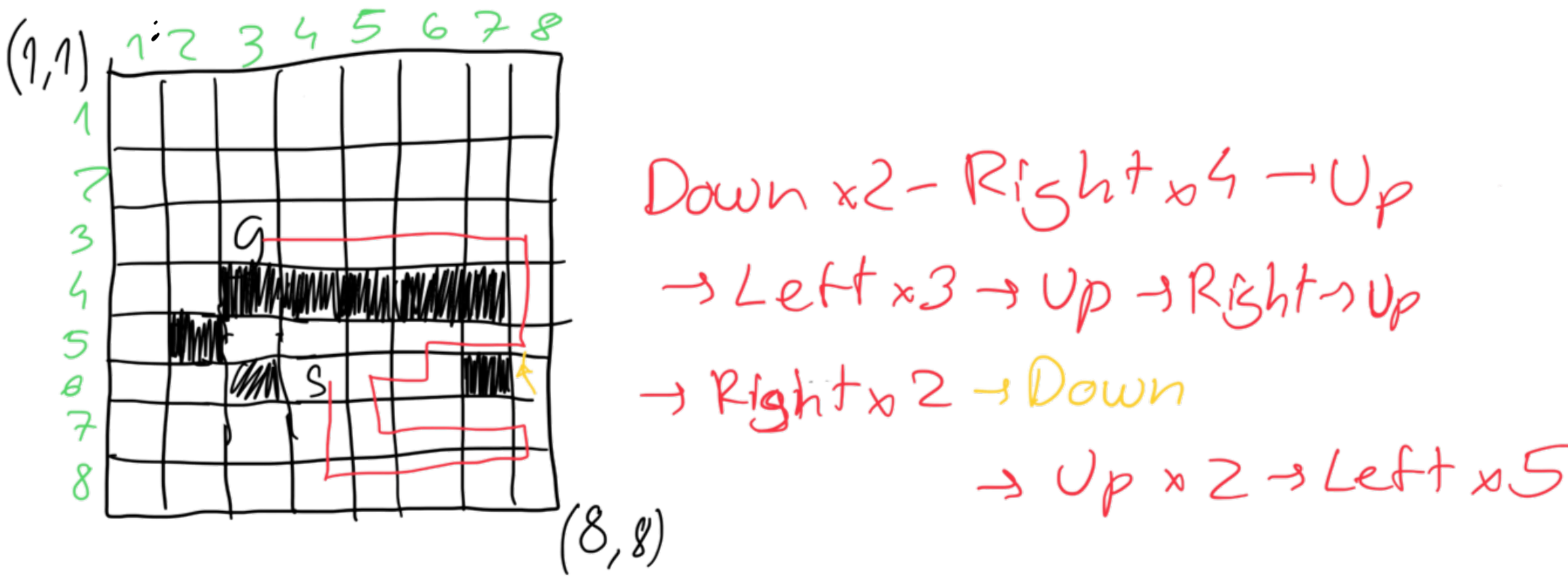


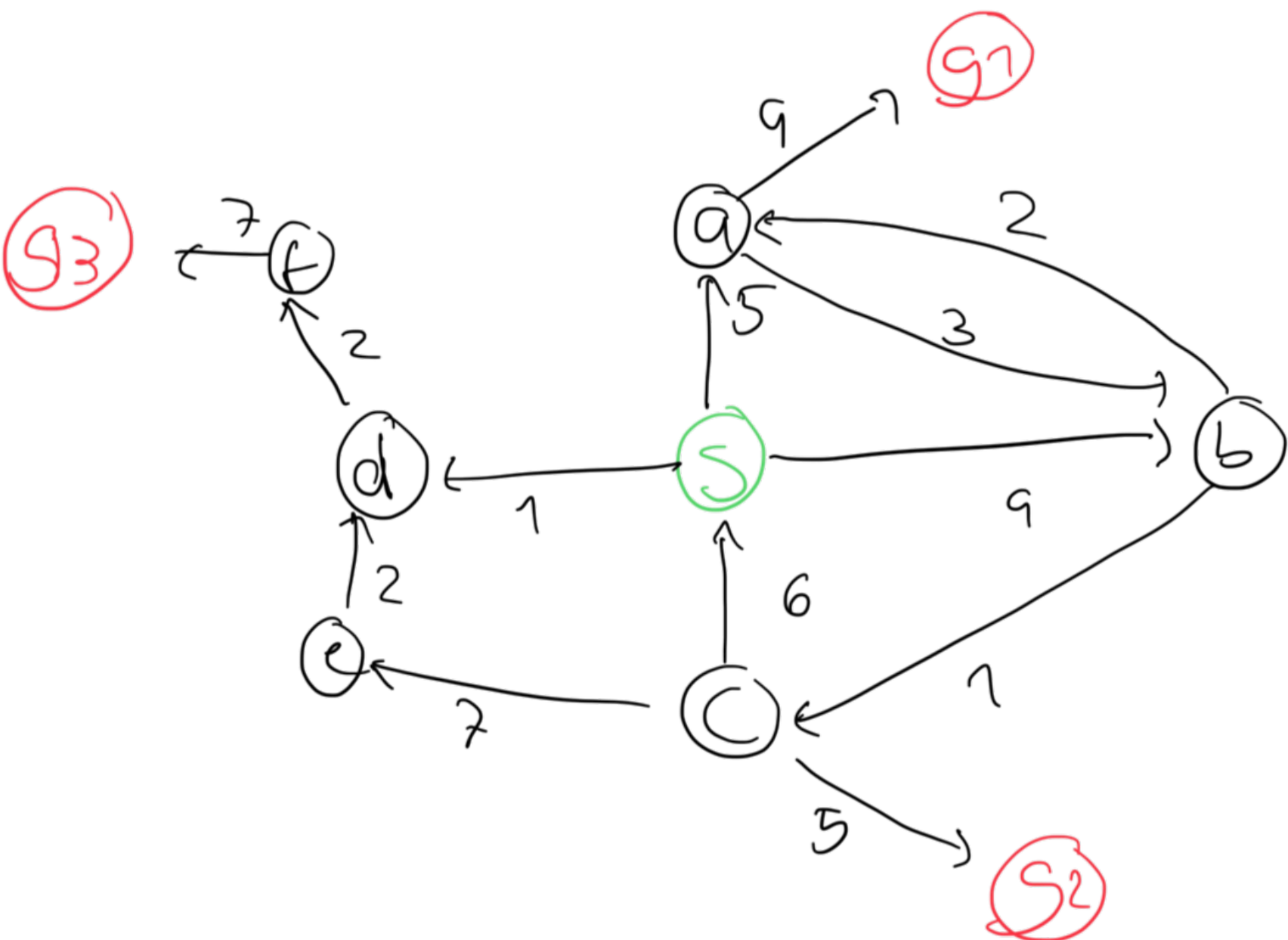
AGK 2

Ü 2.1 Depth-First Search



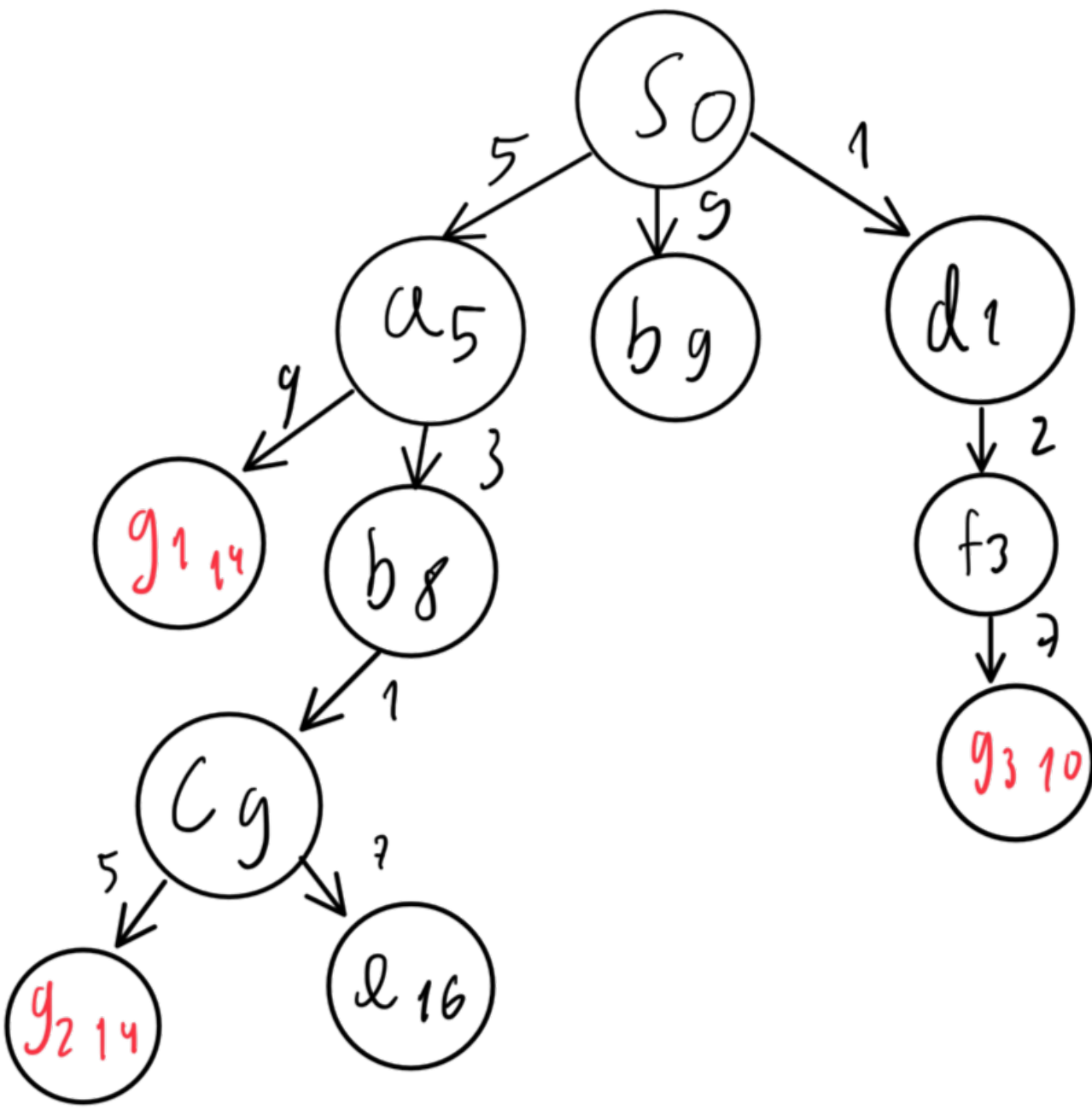
return $S \rightarrow (4,7) \rightarrow (4,8) \rightarrow (5,8) \rightarrow (6,8) \rightarrow (7,8) \rightarrow (8,8)$
 $\rightarrow (8,7) \rightarrow (7,7) \rightarrow (6,7) \rightarrow (5,7) \rightarrow (5,6) \rightarrow (6,6)$
 $\rightarrow (6,5) \rightarrow (7,5) \rightarrow (8,5) \rightarrow (8,4) \rightarrow (8,3) \rightarrow (7,3)$
 $\rightarrow (6,3) \rightarrow (5,3) \rightarrow (4,3) \rightarrow G$

Ü 2.2

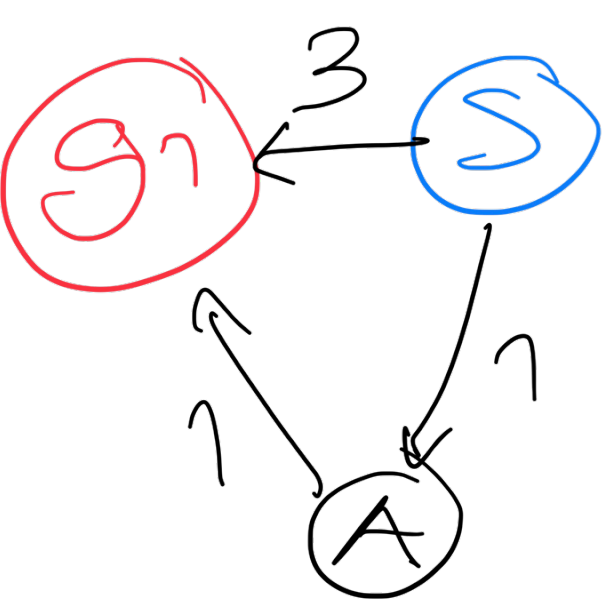


t	Priority Queue	Look up table
0	S:0	S0
1	Sa:5; Sb:9; Sd:1 Sag:14; Sab:8; Sb:9 ; Sbc:10; Sdf:3	S0, a5, b9, d1 S0, a5, b8, d1, c10, f3
2	Sag:14; Sab:9; Sbc:10 ; Sbcg:15 Sbce:17; Sdfg:3:10	S0, a5, b8, d1, c9, f3, g2:15, c17, g3:10
3	Sag:14; Sab:9; Sbcg:14; Sbcce:16; Sdfg:3:10 Sbce:17; Sdfg:3:10	S0, a5, b8, d1, c9, f3, g2:14, c16, g3:10
4	Sbce:17; Sdfg:3:10 Sbce:16; Sdfg:3:10	S0, a5, b8, d1, c9, f3, g2:14, c16, g3:10
5	return Sdfg3:10	

Max. n° of elements in the priority queue: 6



Ü 2.3



It doesn't always return the cost-optimal solution, because it returns the first possible path it finds, and may not be the cost-optimal solution.

Ü 2.4

The priority-queue alone doesn't prevent a node to be explored more than once, with less cost-optimal paths. The look-up table is needed to store the discovered nodes with their respective best discovered costs.